AdifySkills

n8n Automation Mastery
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CLASS NOTES

Module 6 · Production-Grade Engineering

DAY 20 of 25

Scaling, Security & Workflow DevOps

"Ek client se 50 clients tak — bina tootey."

⚡ AAJ HUM YEH BANAYENGE

Nightly Workflow Backup → GitHub

Har raat 2 baje n8n API se saare workflows nikaalta hai aur har ek ko date-wise folder mein GitHub repo mein commit karta hai. Galti se delete ho jaaye to bhi version wapas.

Redis | Queue Mode | GitHub | n8n API

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

💡 Yeh lesson kyun zaroori hai?

Ek client se 50 clients tak jaane ke liye scale, security aur version control chahiye — warna ek galti sab tod deti hai.

🎯 Aaj ke baad aap yeh kar paoge

01 Queue mode with Redis & workers
🌍 Order board + kai chef.
⚙️ EXECUTIONS_MODE=queue, Redis, n8n worker containers.

02 Performance: batching, memory, binary data mode
🌍 Ek baar mein 10 plate, 1000 nahi.
⚙️ Split in Batches, binary data mode = filesystem.

03 Security: credentials, RBAC, secrets, webhook auth
🌍 Ghar ki chaabi sabko nahi dete.
⚙️ Credentials store, user roles, webhook header/basic auth.

04 Version control: export JSON, Git, environments
🌍 Recipe book ki photocopy.
⚙️ n8n API → workflows export JSON → GitHub commit.

05 Testing workflows & staging vs production
🌍 Pehle ghar par try, phir party mein.
⚙️ Staging instance → test data → phir production.

06 Monitoring: execution logs, uptime, metrics
🌍 Dukaan ka CCTV.
⚙️ Execution logs, uptime ping, failed executions count alert.

⏱ Class ka plan (2 ghante)

10m	35m	45m	20m	10m

`0:00-0:10` **Recap**
🪄 *Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.*

`0:10-0:45` **Concept: Scale Architecture**
🪄 *Ek chef 100 log ka khana nahi bana sakta. 5 chef aur ek order board chahiye.*

`0:45-1:30` **Live Build**
Workflow backup to GitHub (auto, nightly)
🪄 *Aaj ki kahani yaad hai — 'Restaurant ka bada kitchen'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.*

`1:30-1:50` **Quiz + Audit**
Module 6 quiz + security audit checklist
🪄 *Practice match — jitna yahan girenge, utna asli match mein kam girenge.*

`1:50-2:00` **Q&A + Homework**
🪄 *Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?*

🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

Queue mode

Main instance + Redis + multiple workers — parallel executions.

🪄 **Analogy:** Restaurant: ek order-taker, kai chefs

CONCEPT 02

Binary data mode

Files memory ki jagah disk/S3 par — RAM bachti hai.

🪄 **Analogy:** Saamaan godown mein rakho, kamre mein nahi

CONCEPT 03

Source control

Workflows JSON ko Git mein — history + rollback.

🪄 **Analogy:** Google Docs version history

CONCEPT 04

Least privilege

Har credential ko sirf zaroori permission.

🪄 **Analogy:** Guest ko sirf drawing room ki chaabi

🔑 Aaj ka main idea

Scale = zyada load sambhalna (queue mode, workers, batching). Security = credentials, access aur webhooks ko surakshit rakhna. DevOps = workflows ka version, testing aur monitoring.

- Redis + 3 workers se 10× throughput
- Webhook par header auth
- Har raat workflows GitHub par backup

📖 Ek kahani se samjho: Restaurant ka bada kitchen

1. 🧾 **Bahut saare orders aaye** = Executions queue
2. 📋 **Order board par lage** = Redis queue
3. 👨‍🍳 **5 chef saath kaam karein** = Workers
4. 🔐 **Kitchen mein sirf staff** = RBAC + credentials
5. 📸 **Har recipe ki copy** = Git version control

🪄 Ek line mein: “Ek chef 100 log ka khana nahi bana sakta. 5 chef aur ek order board chahiye.”

🗺️ Aaj ka workflow: Nightly Workflow Backup → GitHub

Data left se right chalta hai. Har dabba ek n8n node hai.



💰 Isse kitna kama sakte ho?

Agencies with many workflows (backup + DevOps) — aam taur par **₹5,000 – ₹15,000 add-on**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

🛠 Step-by-step build — n8n kholo aur saath chalo

1. ⚡ **Schedule Trigger**
   Roz raat 2 baje
2. 🌐 **n8n (API node)**
   Get all workflows
3. 🔁 **Loop Over Items (Split in Batches)**
   Har workflow ek-ek karke
4. 👨‍💻 **Code**
   JSON pretty-print, file name = workflow-name.json
5. 🐙 **GitHub**
   Create/update file in backup repo
6. ✈️ **Telegram**
   "✅ 37 workflows backed up"

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

🔑 Setup (import ke baad)

- ☐ Connect: **n8n API key (Settings → n8n API)**
- ☐ Connect: **GitHub (personal access token)**
- ☐ GitHub par private repo "n8n-backups" banao
- ☐ n8n Settings → n8n API → API key banao; credential mein Base URL = https://your-n8n-domain/api/v1
- ☐ GitHub token mein "Contents: read & write" permission do

✅ Test kaise karein

1. "Test workflow" → GitHub repo mein backups/<aaj ki date>/ folder check karo

⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

😩 Pehle (manual)	⚡ Ab (n8n)
Ek server, ek line mein sab — slow	Queue mode: workers parallel

⏱ 1000 webhooks/minute bhi bina atke.

⌨️ Aaj ka key snippet

Copy karke n8n mein paste karo 📋 Select karke copy karo

```
EXECUTIONS_MODE=queue
QUEUE_BULL_REDIS_HOST=redis
N8N_DEFAULT_BINARY_DATA_MODE=filesystem
```

📦 Sample data — input se output tak

📥 Input 📋 Select karke copy karo

```
{
  "workflows": 37
}
```

📤 Output 📋 Select karke copy karo

```
{
  "committed": 37,
  "repo": "yourbrand/n8n-backup"
}
```

💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

💬 Build it with ChatGPT / Claude 📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Har raat 2 baje n8n API se saare workflows nikaalta hai aur har ek ko date-wise folder mein GitHub repo mein commit karta hai. Galti se delete ho jaaye to bhi version wapas.

Flow: Schedule (2:00 AM) → n8n API (Saare workflows) → Code (File path + JSON) → GitHub (Commit file)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

💬 Debug helper prompt 📋 Select karke copy karo

```
I'm building the n8n workflow "Nightly Workflow Backup → GitHub" (nodes: Every Night 2 AM, Get All Workflows, Prepare Files, Commit to GitHub).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo (`Ctrl` + `A` , `Ctrl` + `C`).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

⚡ **Workflow JSON**
4 nodes · ready to import
`adifyskills.com/workflows/day-20-nightly-github-backup.json`

🌐 **Lesson page**
Copy buttons, quiz aur video links
`adifyskills.com/day/20`

🧯 Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

Agar yeh ho raha hai…	To yeh karo
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "n8n API key (Settings → n8n API)" connect karo.
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "GitHub (personal access token)" connect karo.
Time par apne aap nahi chala?	"Active" toggle on hai? Settings mein timezone Asia/Kolkata check karo.

⚠️ Yeh galtiyan sabse common hain

- ! Production mein direct edit — staging nahi
- ! API keys workflow JSON mein hardcoded (Git par leak)
- ! Execution data kabhi prune na karna — DB full

⭐ Pro tips — time bachane wali tricks

- ★ EXECUTIONS_DATA_PRUNE=true + max age set karo
- ★ Credentials kabhi export mat karo Git mein
- ★ Uptime monitoring (UptimeRobot) on /healthz

Phir bhi atke? Koi baat nahi 🙂
Screenshot + error message ke saath WhatsApp par poochho →

🏋️ Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

🛠 Live lab

Nightly workflow: saare workflows export karke GitHub repo mein commit karna.

📝 Homework

Apne instance ka security audit checklist ke hisaab se karo aur report submit karo.

- [ ] Homework ho gaya

BEGINNER

Workflow manually export/import

- [ ]

INTERMEDIATE

Nightly GitHub backup

- [ ]

ADVANCED

Staging → production promotion flow

- [ ]

🧩 Aaj ka challenge — pehle khud try karo

🎯 Backup ke baad Telegram par "✅ 12 workflows backed up" message aaye.

💡 Hint 1

Commit ke baad saare items ko gino.

💡 Hint 2

Aggregate ya Code node: return [{json: {count: $input.all().length}}]

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

Q1. Parallel executions ke liye mode?

- (A) Queue mode
- (B) Manual mode
- (C) Test mode
- (D) Pin mode

Q2. Queue mode ko kya chahiye?

- (A) SMTP
- (B) Redis
- (C) MongoDB
- (D) FTP

Q3. Workflows ki history ke liye?

- (A) Email
- (B) Pen drive
- (C) Git / source control
- (D) Screenshots

Q4. DB full hone se bachne ke liye?

- (A) Bada font
- (B) More nodes
- (C) Kuch nahi
- (D) Execution pruning

📖 Aaj ke naye shabd

Workflow
Steps ki ek chain jo apne aap chalti hai — jaise recipe ke steps.

Node
Workflow ka ek dabba/step. Har node ek kaam karta hai (email bhejna, data laana…).

Trigger
Jo workflow ko START karta hai — jaise alarm, naya email, ya form submit.

Webhook
Ek URL jis par koi doosra app data bhejta hai aur aapka workflow turant chal padta hai.

API
Do apps ke baat karne ka tareeka. Ek app sawaal bhejta hai, doosra jawab (data) deta hai.

JSON
Data likhne ka format: { "naam": "Ravi", "city": "Delhi" } — key aur value ke jode.

Credential
Kisi app ka login/API key jo n8n safe tareeke se yaad rakhta hai.

Execution
Workflow ka ek baar chalna. Har run ka record Executions tab mein dikhta hai.

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✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 21

Project Sprint 1: AI Customer Support Desk

"24×7 support jo kabhi chhutti nahi leta."

☐ **Day 20 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

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🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

Q1 → A
Queue mode

Q2 → B
Redis

Q3 → C
Git / source control

Q4 → D
Execution pruning

✅ Challenge ka solution

Commit to GitHub → Code: return [{json: {count: $input.all().length}}] → Telegram "✅ {{ $json.count }} workflows backed up".

📚 Official docs

- Queue mode ↗
- Security ↗
- Source control ↗
- n8n public API ↗

▶ Videos

- n8n queue mode workers redis ↗
- n8n backup workflows github ↗

💬 Doubt poochho
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wa.me/918383891716

🌐 Day 20 lesson page
Quiz, copy buttons, videos
adifyskills.com/day/20

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